Designing the Vertical System of Record

The Ecosystem, Chapter 7/9 — Revision & Viva-Prep Cheatsheet (English + Roman Urdu)
Source: agentfactory.panaversity.org/docs/ecosystem-designing-the-vertical-sor

EN: A first-principles, legacy-informed redesign method. Most of an old workflow isn't the profession itself -- it's a workaround for human limits (scarce attention, scattered info, department handoffs) or old technology. Sort every element into 3 bins, then rebuild reflexes from scratch -- for AI Workers, not humans.
Roman Urdu: First-principles, legacy-informed redesign method. Purana workflow zyada tar profession nahi -- human limits (scarce attention, scattered info, department handoffs) ya purani technology ka workaround hai. Har element teen bins mein sort karo, phir reflexes scratch se rebuild karo -- AI Workers ke liye, insaano ke liye nahi.

1. First Principles + Workflow Archaeology

- 5 pre-agentic limits shaped every old workflow: scarce attention, scattered info, software that can't understand, department divisions, manager blindness.
- PwC 2026: 56% of CEOs got no financial return from AI -- not model quality, but skipped fundamentals (clean data, sound process, governance).
- Start from the outcome, then dig into 5 real cases with the expert: normal, difficult, failed, escalated, expert-catch.

2. Three-Bin Sort + Source Hierarchy

- Ask of every element: why does this exist? **Bin 1** (law/trust -- keep), **Bin 2** (human limits -- redesign), **Bin 3** (old tech -- delete). When in doubt, Chesterton's Fence: keep in Bin 1 until the expert confirms otherwise.
- Source hierarchy: a 7-rung typical ladder -- relevance alone isn't enough, a source must also be applicable (e.g. right jurisdiction).

3. Two Content Classes + Decisions/Rules/Judgment

- **Authority** (cited, Worker-facing, tested) vs. **Orientation** (short, human-facing, never cited).
- Map **decisions**, not departments. **Rules** (automatic checks) vs. **Judgment** (tested interpretation, never hidden in a prompt) vs. **Permissions** (Read -> Execute-irreversible are separate grants).
- Exceptions are designed BEFORE the normal path -- that's what actually reduces human escalation work.

4. Eight Failure Modes + Templates

- Document dump, AI-readable SOP, technology-first, rules hidden in prompts, authority-before-evidence, happy-path demo, first-principles theater, unmarked textbook.
- **7 Templates:** Outcome Contract, Sort Record, Invariants List, Source Register, Decision Map, Exception Entry, Coverage Register.

Key Terms Cheatsheet

One line each — English + Roman Urdu.

Term	English	Roman Urdu
Three-Bin Sort	Law/trust (keep), human limits (redesign), old tech (delete).	Law/trust (rakho), human limits (redesign), old tech (delete).
Chesterton's Fence	When in doubt, keep an element in Bin 1 until the expert confirms it can move.	Shak ho to Bin 1 mein rakho jab tak expert confirm na kare.
Authority vs Orientation content	Cited Worker-facing rules vs. short human-facing context, never cited.	Cited Worker-facing rules vs. short human-facing context, kabhi cited nahi.
Thin slice	One professional outcome fully covered -- the credential that earns a sponsor conversation.	Ek professional outcome poori tarah covered -- jo sponsor conversation kamata hai.
Coverage Register	A template recording what's complete as fact, not feeling.	Template jo completeness ko fact ki tarah record karta hai, feeling nahi.

Quick Self-Check

Question	Answer
Why isn't the old workflow a good blueprint for an AI Worker?	Most of it is a workaround for human limits or old tech, not the profession itself -- redesigning from first principles separates the two.
What's the test for whether something belongs in Bin 1?	Does it exist for law/trust reasons? If in doubt, Chesterton's Fence keeps it in Bin 1 until the expert confirms otherwise.
Why map decisions instead of departments?	A customer can assign any decision to any team -- departments are organizational accident, decisions are the real unit of work.
Why are exceptions designed before the normal path?	Well-designed exception handling is what actually reduces the human escalation workload -- it's not an afterthought.

Where This Leads / Sources

Leads to: AI Searchable Context, Skills & Connectors, Spec-Driven Development, Eval-Driven Development, building the first domain Worker. Full 59-Q quiz in 10-test-your-understanding.md; practice project in projects/fill-the-templates/.

Source: docs/ecosystem-designing-the-vertical-sor/ (this repo), content originally via Zia Tutor AI connector.